The Very Basic Note in Number Theory

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Contents

1 Divisibilities

Theorem 1.1. (Division algorithm(除法算則)) Let a and b be integers where $b \neq 0$. Then there exist unique integers q and r such that

$$a = qb + r \quad \text{and} \quad 0 \leqslant r \leqslant |b| - 1.$$

This is simply a formal way to express the division of an integer by a nonzero integer. Here, q is called the *quotient*(商) while r is called the *remainder*(餘數) when a is divided by b.

Definition 1.1. An (nonzero) integer b is said to *divide*(整除) an integer a if $a = kb$ for some integer k. We use the notation $b \mid a$ to denote this. If there is no such integer k, then b does not divide a, and we use the notation $b \nmid a$.

If $b \mid a$, we also say that b is a *divisor*(因子) (or 約數 in CMO and CGMO) of a, and a is a *multiple*(倍數) of b, or a is *divisible by*(被...整除) b. Note that these definitions apply to negative integers as well. For example, we have $-2 \mid 4$ and $-3 \nmid -4$. By convention, we usually assume b is nonzero when we write $b \mid a$ or $b \nmid a$. However, some may take into account the case $b = 0$ in the above definition. This means $0 \mid a$ if and only if $a = 0$.

Proposition 1.1. Let a, b, c, d be integers. Then the following hold.

(a) If $a \mid b$ and $b \mid c$, then $a \mid c$.

(b) If $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ for any integers m and n.

(c) If $a \mid b$ and $c \mid d$, then $ac \mid bd$.

(d) If $a \mid b$ and $b \neq 0$, then $|a| \leqslant |b|$.

Proof. See proposition 1.1 of **Elementary Number Theory 3.1**. □

Definition 1.2. Let a and b be integers which are not both zero. A nonzero integer c is said to be a *common divisor*(公因數) of a and b if $c \mid a$ and $c \mid b$. In particular, there exists a unique integer $d > 0$, called the *greatest common divisor*(最大公因數) of a and b, such that

(i) $d \mid a$ and $d \mid b$, and

(ii) whenever $c \mid a$ and $c \mid b$, we have $c \leqslant d$.

The greatest common divisor (also known as 最大公約數 in CMO and CGMO) of a and b is denoted by $d = \gcd(a,b)$ or simply $d = (a,b)$. In particular, if $(a,b) = 1$, we say that a and b are *relatively prime*(互質) (or 互素 in CMO and CGMO) or *coprime*.

Example 1.1. We have $(6,9) = 3$, $(4,-6) = 2$ and $(-15,-20) = 5$.

Proposition 1.2. Let a,b,c be integers. Then the following hold.

(a) $(a,b) = (a+kb,b)$ for any integer k.

(b) $(ka,kb) = |k|\,(a,b)$ for any nonzero integer k.

(c) If $(a,c) = 1$, then $(a,bc) = (a,b)$.

(d) If $a \mid bc$, then $\dfrac{a}{(a,c)} \mid b$. In particular, if $a \mid bc$ and $(a,c) = 1$, then $a \mid b$.

(e) (Bézout's identity(貝祖定理)) If $d = (a,b)$, then there exist integers m and n such that $ma + nb = d$.

(f) If $c \mid a$ and $c \mid b$, then $c \mid (a,b)$.

Proof. See proposition 1.5 of **Elementary Number Theory 3.1**. □

Example 1.2. To find $m,n \in \mathbb{Z}$ such that $27m + 155n = 1$ (where $(27,155) = 1$), we may apply the *Euclidean algorithm*(輾轉相除法) as follows:

$$\begin{aligned}
155 &= 5 \times 27 + 20,\\
27 &= 1 \times 20 + 7,\\
20 &= 2 \times 7 + 6,\\
7 &= 1 \times 6 + 1.
\end{aligned}$$

Using these relations, we get

$$\begin{aligned}
1 &= 7 - 1 \times 6\\
&= 7 - 1 \times (20 - 2 \times 7)\\
&= -1 \times 20 + 3 \times 7\\
&= -1 \times 20 + 3 \times (27 - 1 \times 20)\\
&= 3 \times 27 - 4 \times 20\\
&= 3 \times 27 - 4 \times (155 - 5 \times 27)\\
&= -4 \times 155 + 23 \times 27.
\end{aligned}$$

Thus, we may take $m = 23$ and $n = -4$. However, such a pair of integers is not unique.

Proposition 1.3. Let a, b, c be integers. Then $am + bn = c$ has integer solutions in m, n if and only if $d \mid c$ where $d = (a, b)$. In that case, if $(m, n) = (m_0, n_0)$ is a solution, then all solutions are given by

$$m = m_0 + \frac{b}{d} \cdot k \quad \text{and} \quad n = n_0 - \frac{a}{d} \cdot k$$

where k is any integer.

Proof. See proposition 2.1 of **Elementary Number Theory 3.1**. □

Definition 1.3. Let a and b be nonzero integers. An integer c is said to be a *common multiple*(公倍數) of a and b if $a \mid c$ and $b \mid c$. In particular, there exists a unique integer $\ell > 0$, called the *least common multiple*(最小公倍數) of a and b, such that

(i) $a \mid \ell$ and $b \mid \ell$, and

(ii) whenever $a \mid c$ and $b \mid c$ for $c \neq 0$, we have $\ell \leqslant |c|$.

The least common multiple of a and b is denoted by $\ell = \mathrm{lcm}(a, b)$ or simply $\ell = [a, b]$.

Example 1.3. We have $[6, 9] = 18$, $[4, -6] = 12$ and $[-3, -5] = 15$.

Proposition 1.4. Let a, b, c be integers. Then the following hold.

(a) $|a|\,|b| = (a, b)[a, b]$

(b) if $a \mid c$ and $b \mid c$, then $[a, b] \mid c$.

Proof. See proposition 1.7 of **Elementary Number Theory 3.1**. □

In particular, if $(a, b) = 1$, then $[a, b] = |a|\,|b|$. Also, if $a \mid c$, $b \mid c$ and $(a, b) = 1$, then $ab \mid c$.

2 Prime Numbers

Definition 2.1. A positive integer $p > 1$ is called a *prime number*(質數) or simply a *prime* if its only positive divisors are 1 and p. A positive integer $n > 1$ which is not a prime number is called a *composite number*(合成數).

In CMO and CGMO, a prime number is also known as 素數. Note that 1 is neither a prime nor a composite number.

Example 2.1. The first few primes are $2, 3, 5, 7, 11, \ldots$, while the first few composite numbers are $4, 6, 8, 9, 10, \ldots$. It is known that there are infinitely many primes and infinitely many composite numbers.

Theorem 2.1. (Fundamental theorem of arithmetic(算術基本定理)) Every positive integer can be uniquely expressed as a product of distinct prime powers up to order.

Proposition 2.1. Let p be a prime number, let a, b be integers, and let n be a positive integer. Then the following hold.

(a) If $a \mid p$, then $a = \pm 1, \pm p$.

(b) If $p \mid ab$, then $p \mid a$ or $p \mid b$.

(c) If $p \mid a^n$, then $p \mid a$.

(d) If $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ and $b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ where $p_1, p_2, \ldots, p_s$ are distinct primes, then

$$(a, b) = p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s} \quad \text{and} \quad [a, b] = p_1^{\delta_1} p_2^{\delta_2} \cdots p_s^{\delta_s}$$

where $\gamma_j = \min\{\alpha_j, \beta_j\}$ and $\delta_j = \max\{\alpha_j, \beta_j\}$ for $j = 1, 2, \ldots, s$.

Proof. See propositions 1.3, 1.4 and 1.6 of **Elementary Number Theory 3.1**. □

Definition 2.2. Let α be a real number, and let n be a positive integer. We define a *divisor function*(除數函數) of n by

$$\sigma_\alpha(n) = \sum_{d \mid n} d^\alpha,$$

where the summation runs through all positive divisors d of n.

In particular, when $\alpha = 0$, $\sigma_0(n)$ is the number of positive divisors of n and is usually denoted by $d(n)$ or $\tau(n)$. When $\alpha = 1$, $\sigma_1(n)$ is the sum of all positive divisors of n and is usually denoted by $\sigma(n)$.

Proposition 2.2. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be the prime factorization of $n > 1$. Then the following hold.

(a) $d(n) = (\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1)$

(b) $\sigma(n) = (1 + p_1 + \cdots + p_1^{\alpha_1})(1 + p_2 + \cdots + p_2^{\alpha_2}) \cdots (1 + p_s + \cdots + p_s^{\alpha_s})$

Proof. See proposition 1.8 of **Elementary Number Theory 3.1**. □

Example 2.2. The positive divisors of $12 = 2^2 \cdot 3$ are $1, 2, 3, 4, 6, 12$. Therefore, we have $d(12) = 6$ and $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$. These agree with the formulae $d(12) = (2 + 1)(1 + 1)$ and $\sigma(n) = (1 + 2 + 4)(1 + 3)$ given by proposition 2.2.

Definition 2.3. Let n be a positive integer. We define $\varphi(n)$ to be the number of positive integers from 1 to n which are coprime to n. The function φ is called *Euler's totient function* or simply the *Euler function*(歐拉函數).

Proposition 2.3. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be the prime factorization of $n > 1$. Then

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_s}\right).$$

Proof. See proposition 1.9 of **Elementary Number Theory 3.1**. □

In particular, for any prime number p and any integer $k \geqslant 1$, we have $\varphi(p) = p - 1$ and $\varphi(p^k) = p^{k-1}(p - 1)$.

Example 2.3. For $n = 12$, note that only $1, 5, 7, 11$ are coprime to 12 in the range from 1 to 12. Therefore, we have $\varphi(12) = 4$ by definition. This agrees with the formula $\varphi(12) = 12\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 4$ given by proposition 2.3.

3 Congruences

Definition 3.1. Let n be a positive integer. If a and b leave the same remainder when divided by n, then we say that a is *congruent*(同餘) to b modulo n, or symbolically,

$$a \equiv b \pmod{n}.$$

Note that $a \equiv b \pmod{n}$ if and only if $n \mid a - b$. If a is not congruent to b modulo n, then we write $a \not\equiv b \pmod{n}$.

Example 3.1. We have $9 \equiv 5 \pmod{4}$ and $2 \equiv -3 \pmod{5}$, but $-7 \not\equiv 9 \pmod{3}$.

Proposition 3.1. Let a, b, c, d, m, n be integers where m, n are positive. Then the following hold.

(a) $a \equiv a \pmod{n}$

(b) if $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$

(c) if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$

(d) if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a \pm c \equiv b \pm d \pmod{n} \quad \text{and} \quad ac \equiv bd \pmod{n}$$

(e) if $a \equiv b \pmod{n}$ and $c \mid a$, $c \mid b$, then $\frac{a}{c} \equiv \frac{b}{c} \left(\bmod \frac{n}{(n,c)}\right)$

(f) if $a \equiv b \pmod{m}$ and $a \equiv b \pmod{n}$, then $a \equiv b \pmod{[m,n]}$

Proof. See proposition 1.10 of **Elementary Number Theory 3.1**. □

These properties suggest the congruence relation behaves almost the same as the usual equality, except that we need to be careful when doing division, as can be seen from part (e). Here, we list out some special cases of the above facts. As a corollary of (d), if $a \equiv b \pmod{n}$ and $k \in \mathbb{Z}$, then

$$\begin{aligned} a \pm k &\equiv b \pm k \pmod{n}, \\ ka &\equiv kb \pmod{n}, \\ a^k &\equiv b^k \pmod{n} \quad \text{for } k \geqslant 0. \end{aligned}$$

Thus, if $P(m)$ is a polynomial with integer coefficients, then $P(a) \equiv P(b) \pmod{n}$. This implies $a - b \mid P(a) - P(b)$.

For (e), if $a \equiv b \pmod{n}$ and $(n,c) = 1$, then $\dfrac{a}{c} \equiv \dfrac{b}{c} \pmod{n}$.

For (f), if $a \equiv b \pmod{m}$ and $a \equiv b \pmod{n}$, and $(m,n) = 1$, then $a \equiv b \pmod{mn}$.

Definition 3.2. Let n be a positive integer, and let $P(m)$ be a polynomial with integer coefficients. An integer a is called a *solution*(解) or a *root*(根) to the *congruence equation*(同餘方程) $P(m) \equiv 0 \pmod{n}$ if $P(a) \equiv 0 \pmod{n}$.

Suppose a is a solution to $P(m) \equiv 0 \pmod{n}$. If $b \equiv a \pmod{n}$, then b is also a solution to the equation from the above observation. Therefore, they are regarded as the same solution. Two solutions a and b are said to be distinct if $a \not\equiv b \pmod{n}$.

Proposition 3.2. Let a, b be integers and let n be a positive integer. Then the congruence equation $ax \equiv b \pmod{n}$ in x is solvable if and only if $(a,n) \mid b$.

Proof. See proposition 1.11 of **Elementary Number Theory 3.1**. $\square$

Example 3.2. To solve $5x \equiv 2 \pmod{17}$, we list out those integers congruent to 2 modulo 17, which are $2, 19, 36, 53, 70, \ldots$. Thus, we get $5x \equiv 70 \pmod{17}$, and hence $x \equiv 14 \pmod{17}$ as $(5,17) = 1$.

Proposition 3.3. Let n be an integer. Then the following hold.

(a) $n^2 \equiv 0, 1 \pmod{3}$

(b) $n^2 \equiv 0, 1 \pmod{4}$

(c) $n^2 \equiv 0, 1, 4 \pmod{5}$

(d) $n^2 \equiv 0, 1, 4 \pmod{8}$

Proof. For example, to find all possible remainders of square numbers modulo 8, it suffices to compute

$$0^2, 1^2, \ldots, 7^2 \equiv 0, 1, 4, 1, 0, 1, 4, 1 \pmod{8},$$

since every integer is congruent to one of $0, 1, \ldots, 7 \pmod{8}$. Indeed, by noticing $a^2 \equiv (-a)^2 \pmod{8}$, it suffices to consider $0^2, 1^2, \ldots, 4^2$. $\square$

Proposition 3.4. Let p be a prime number. Define

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

where $a_n, a_{n-1}, \ldots, a_0$ are integers. If $p \nmid a_n$, then $f(x) \equiv 0 \pmod{p}$ has at most n distinct solutions modulo p.

Proof. See proposition 4.1 of **Elementary Number Theory 3.1**. □

Example 3.3. It is obvious that $x \equiv 2 \pmod{113}$ and $x \equiv 3 \pmod{113}$ are solutions to the quadratic congruence equation $x^2 - 5x + 6 \equiv 0 \pmod{113}$. Since 113 is a prime, proposition 3.4 ensures that these are the only 2 solutions.

Example 3.4. The congruence equation $x^2 \equiv 1 \pmod 8$ has exactly 4 solutions $x \equiv 1, 3, 5, 7 \pmod 8$. This shows that the result in proposition 3.4 may not be true if p is not a prime.

Theorem 3.1. (Chinese remainder theorem(中國剩餘定理)) Let k be a positive integer. Consider a system of congruence equations

$$\begin{cases} x \equiv a_1 \pmod{n_1}, \\ x \equiv a_2 \pmod{n_2}, \\ \quad \cdots \\ x \equiv a_k \pmod{n_k} \end{cases}$$

in x where $n_1, n_2, \ldots, n_k$ are pairwise coprime positive integers, and $a_1, a_2, \ldots, a_k$ are integers. Then there exists a unique solution modulo $n_1 n_2 \cdots n_k$.

Proof. See theorem 1.4 of **Elementary Number Theory 3.1**. □

Example 3.5. To solve

$$\begin{cases} x \equiv 2 \pmod 3, \\ x \equiv 3 \pmod 5, \end{cases}$$

we list out those integers congruent to 3 modulo 5, which are $3, 8, \ldots$. Among these, note that $8 \equiv 2 \pmod 3$. Therefore, 8 is a solution to both congruence equations. By the Chinese remainder theorem, the unique solution is $x \equiv 8 \pmod{15}$.

Theorem 3.2. (Euler-Fermat theorem(歐拉費馬定理)/Euler's theorem(歐拉定理)) Let n be a positive integer, and let a be an integer with $(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod n.$$

Here, $\varphi(n)$ is Euler's totient function.

Proof. See theorem 1.2 of **Elementary Number Theory 3.1**. □

In particular, when n is a prime number, we have the following theorem.

Theorem 3.3. (Fermat little theorem(費馬小定理)) Let p be a prime number, and let a be an integer with $(a, p) = 1$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Sometimes it is more convenient to use the forms

$$a^p \equiv a \pmod{p} \quad \text{and} \quad a^{n+(p-1)d} \equiv a^n \pmod{p}$$

for any nonnegative integers n and d. Note that these also hold if $p \mid a$ unless $n = 0$.

Example 3.6. Since $100 \equiv 4 \pmod{16}$, by the Fermat little theorem, we have

$$3^{100} \equiv 3^4 = 81 \equiv 13 \pmod{17}.$$

Links

Theorems

- Bézout's identity
- Chinese remainder theorem
- Division algorithm
- Euclidean algorithm
- Euler-Fermat theorem
- Euler's theorem
- Fermat little theorem
- Fundamental theorem of arithmetic

Propositions

- 1.1: Properties of divisibilities
- 1.2: Properties of greatest common divisors
- 1.3: Solutions to linear Diophantine equations
- 1.4: Properties of least common multiples
- 2.1: Properties of primes and divisibilities
- 2.2: Formulae of divisor functions
- 2.3: Formula of Euler's totient function
- 3.1: Properties of congruences
- 3.2: Solutions to linear congruence equations
- 3.3: Quadratic residues modulo small integers
- 3.4: Polynomial congruence equation modulo p

Terminologies and Notations

- $a \nmid b$
- $d(n)$
- $\tau(n)$
- $\sigma(n)$
- $a \not\equiv b \pmod{n}$
- common divisor
- common multiple

- composite number
- congruence equation
- congruent $a \equiv b \pmod{n}$
- coprime
- divide $a \mid b$
- divisible by
- divisor
- divisor function $\sigma_\alpha(n)$
- Euler function $\varphi(n)$
- Euler's totient function $\varphi(n)$
- greatest common divisor $\gcd(a, b)$, (a, b)
- least common multiple $[a, b]$
- multiple
- not congruent $a \not\equiv b \pmod{n}$
- not divide $b \nmid a$
- prime number
- quotient
- relatively prime
- remainder
- root (congruence equation)
- solution (congruence equation)